Test report 20 – UI rework

Implementation of the approved concept `docs/konzepte/ui-ueberarbeitung/konzept.html` (all proposals option A, accent teal instead of blue) · Browser and Tauri · 2026-10-24 · Branch worktree-agent-a556826f13eb0d980 · Commit 18b43b7 · 4 cores

Playwright (full suite)	new among them U01–U09	Tauri E2E	Vitest	tsc
140 / 140 0 skipped, 0 flaky	**9** bars, menus, tabs, modes, cube, side panel, keyboard, themes	**18 / 18** incl. T18 (new)	**238** incl. contrast light/dark	**0 errors** Rust cad-app: 1 test, clippy ok, fmt ok

1. What changed

- **Building blocks** in `web/src/app/ui/`: `Button` (primary, standard, quiet, toggle with `aria-pressed`), `IconButton`, `Segmented` (radio group with arrow keys), `MenuButton`/`SplitButton`/`ContextMenu` (WAI-ARIA menus with submenus, rendered into `<body>`, never clipped), `Toolbar`/`ToolGroup`/`ToolSep`. One height 26 px, radius 6 px, Inter 12 px (the Inter 4.1 file shipped with `cad-font` – same widths in Chromium and WebKitGTK). Icons from Lucide (`lucide-react` 1.52, ISC), three custom ones in the same stroke (line, arc, ⊥). Tooltips with keyboard shortcut, `aria-label` = plain name, `aria-keyshortcuts`.
- **Header bar**: app name, "File ▾" (German UI: „Datei ▾“; New, Open, Open folder, Recently opened ›, Save as, Add file, Load file, Load/save ZIP), "Examples ▾" with submenus, file name centered (change dot in the desktop app, path into the module context as previously in the path bar), "Save" primary, split export button (remembers the format, Ctrl+E), theme and settings icon buttons.
- **Editor**: only tabs and "+"; right click/Shift+F10 on a tab: "Set as main file", "Close".
- **View**: the rows "File ..." and "Parts ..." are gone. Toolbar at the top: on the left mode Model/Section/Sketch/Measure (exactly one), on the right Fit, View ▾ (Iso, Top, Front, Right, Back, Left, Bottom), All grids, Display ▾ (grid, color regions, side panel). The mode's context bar always at the top left below it. Clickable view cube at the bottom right instead of the axis triad.
- **Context bars**: Section (X/Y/Z/View, position, flip, copy CLI), Sketch (icon groups, Arc ▾, keyboard shortcuts, "Done" primary), Measure (measurement kind as segmented control, "In section", copy, delete, "Done").
- **Side panel** on the right, collapsible, tabs Parts (visibility, volume, export in the ⋯ menu, "without part", total), Colors, Measure, in the module context Parameters.
- **Accent teal**: light `#0f766e`, dark `#3cc4a8`; likewise focus ring, selection/hover of the view (`--view-accent`, `--view-flat`, `--hover-line`, `--viewhover-line`); the model palette is unchanged.
- **Arch package/Wayland**: the desktop file is named after the app ID (`dev.controller-cad.desktop.desktop`), the app sets the app ID as GLib program name (Wayland `app_id`, X11 `WM_CLASS`).

2. Header bar

Before: ten equally weighted text buttons, two select lists and three export buttons; at 1280 and 1440 px the bar extended beyond the window (it scrolled horizontally, "Settings" and the theme were clipped):

Before (width, theme)	scrollWidth	clientWidth	Result
1280-light	1431	1280	clipped / scrolls
1440-light	1440	1440	fits
1920-light	1920	1920	fits
1280-dark	1431	1280	clipped / scrolls
1440-dark	1440	1440	fits
1920-dark	1920	1920	fits



Before: header bar at 1280 px, light



After: header bar at 1280 px, light



Before: header bar at 1440 px, dark



After: header bar at 1440 px, dark

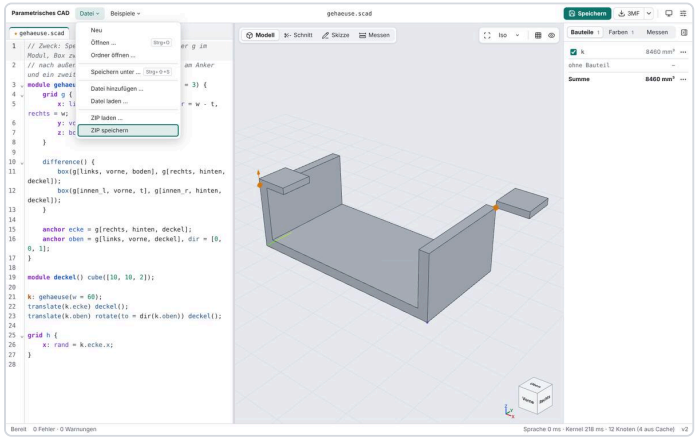
Before: header bar at 1920 px, light



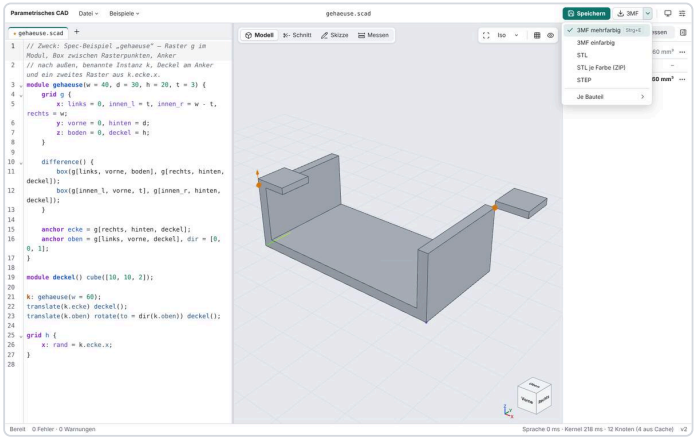
After: header bar at 1920 px, light

U02 Header bar

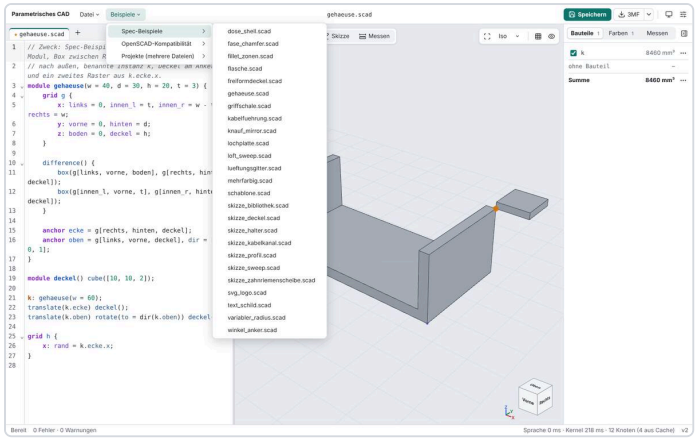
"File ▾" opens with Enter, arrow keys and Home/End move the focus, Esc closes and returns focus to the button. "Examples ▾" with submenus. The split export button lists 3MF multicolor/single color, STL, STL per color, STEP and "Per part"; after choosing STL the button shows "STL", Ctrl+E exports STL, also after reloading. Theme as icon button with menu.



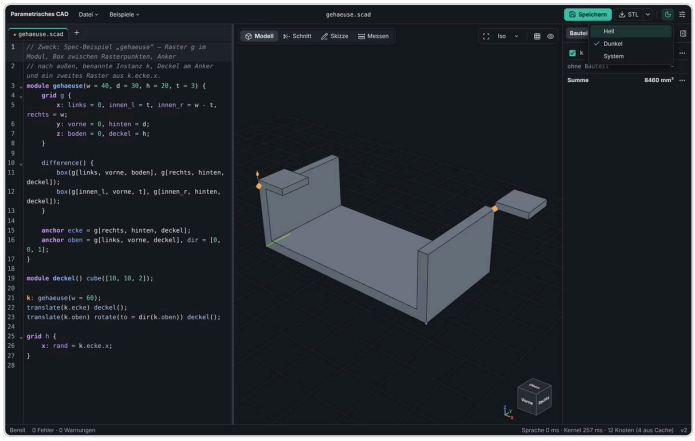
File menu (keyboard)



Export menu



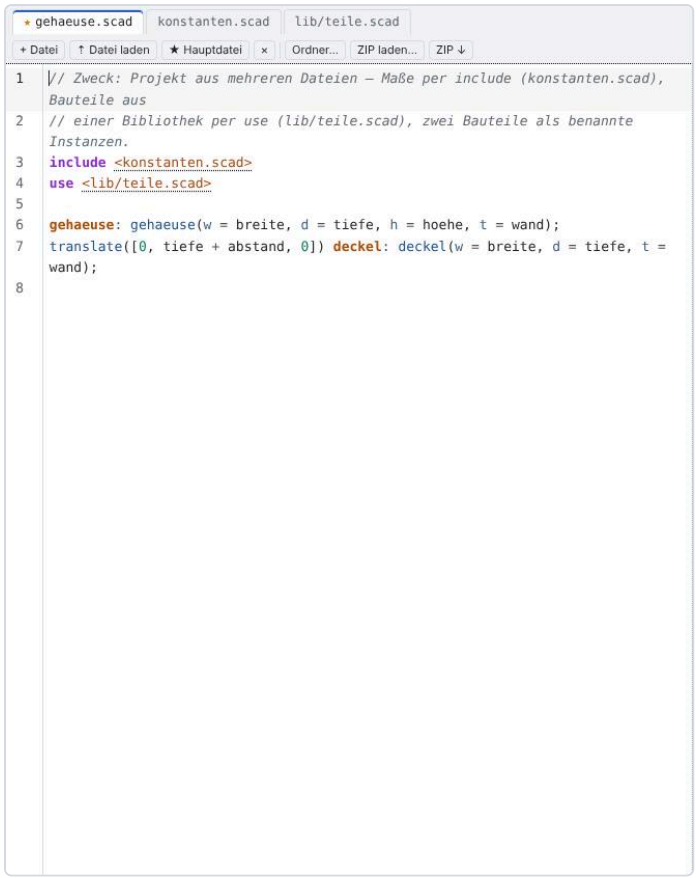
Examples with submenu



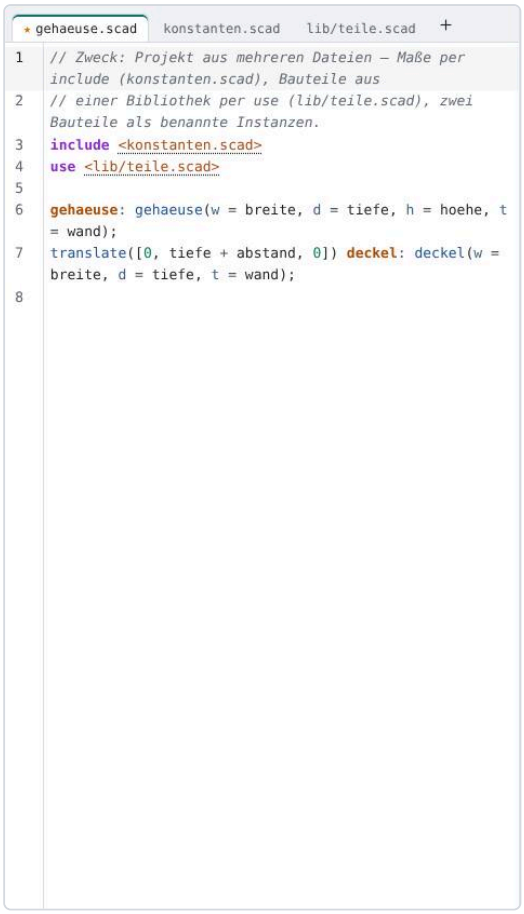
Theme

- 8 entries in the File menu
- export STL via menu and via Ctrl+E (gehaeuse.stl)
- format kept after reload

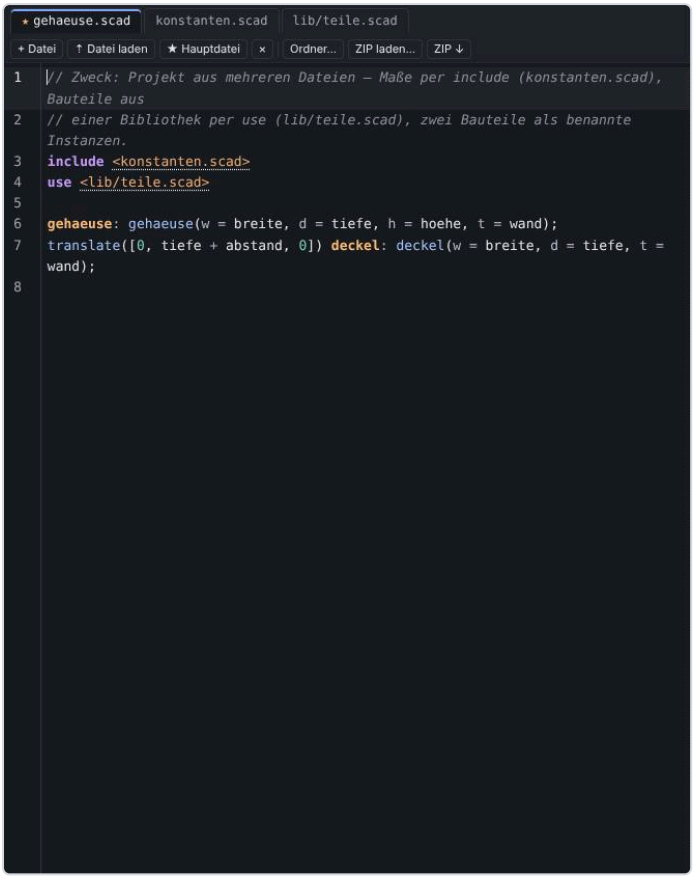
3. Editor



Before: editor with the example project (three files), light



After: editor with the example project (three files), light



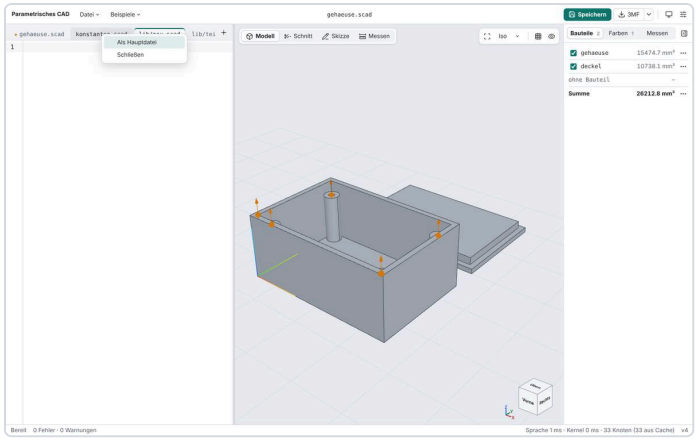
Before: the same, dark



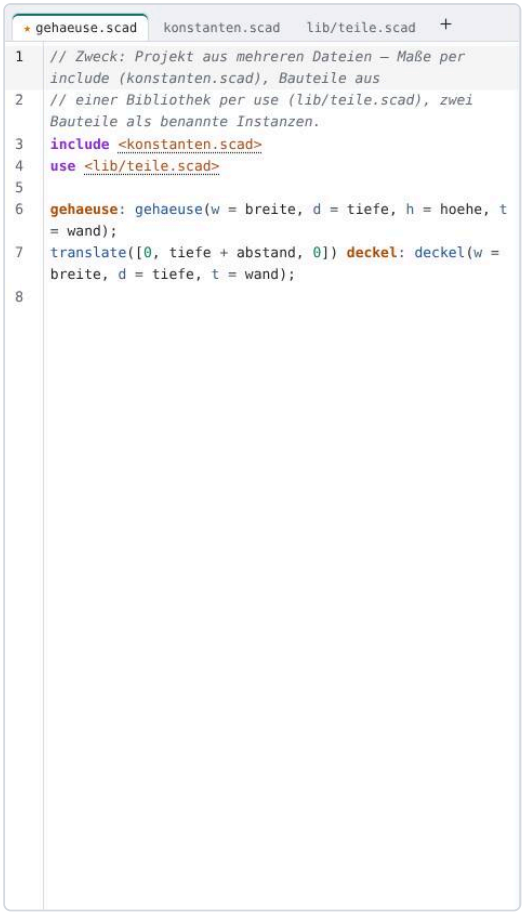
After: the same, dark

U03 Editor tabs

Above the editor only the tabs and "+" (New file, Load file, actions of the displayed file). Right click on a tab: "Set as main file", "Close" (disabled for the main file); via keyboard Shift+F10 or the context menu key. Renaming does not exist (yet).



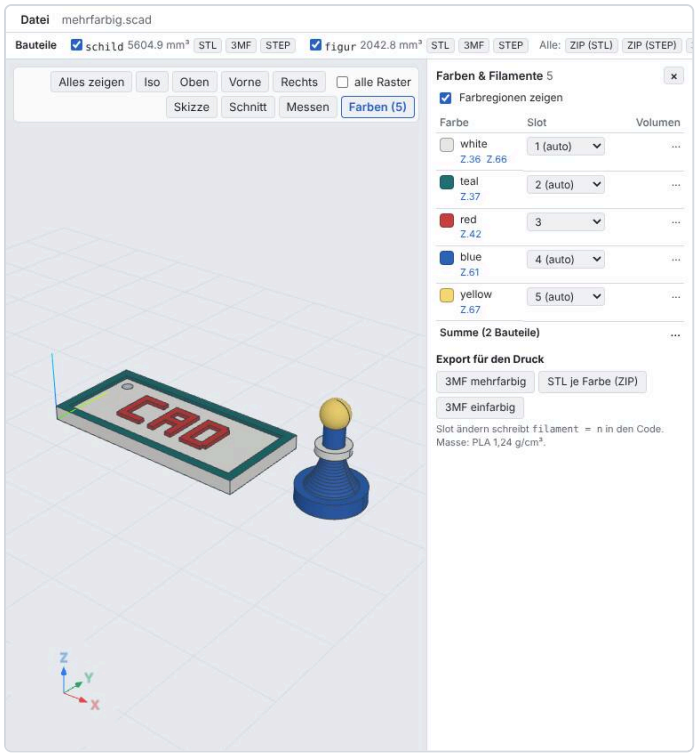
context menu of a tab



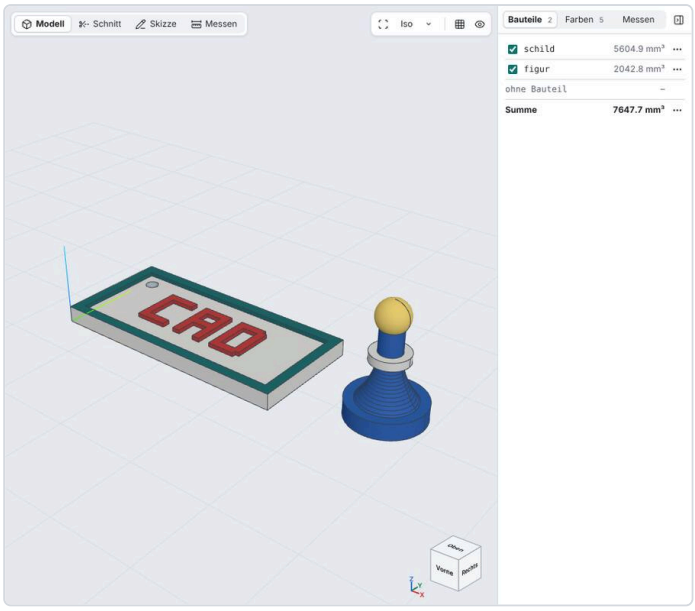
editor with tabs and "+"

- new file via "+"
- main file switched via context menu
- close via keyboard (Shift+F10)

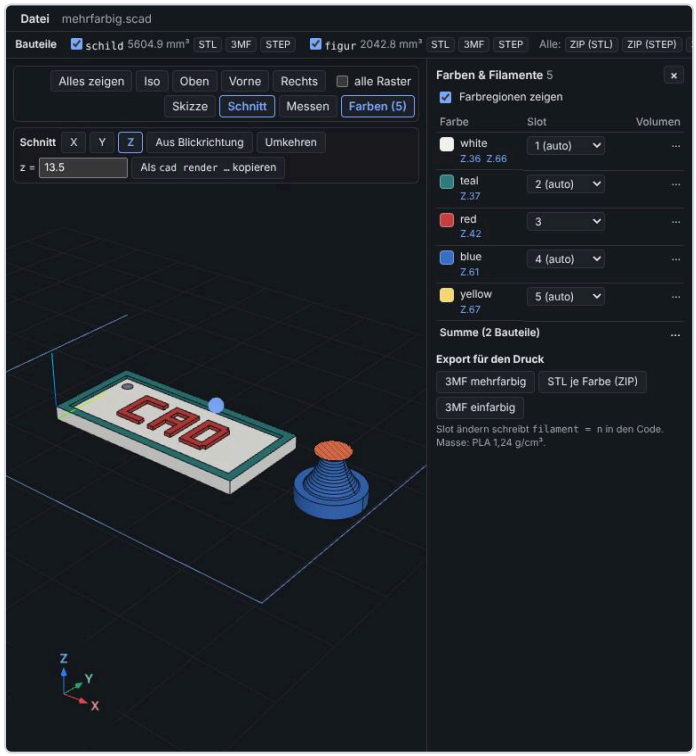
4. View and modes



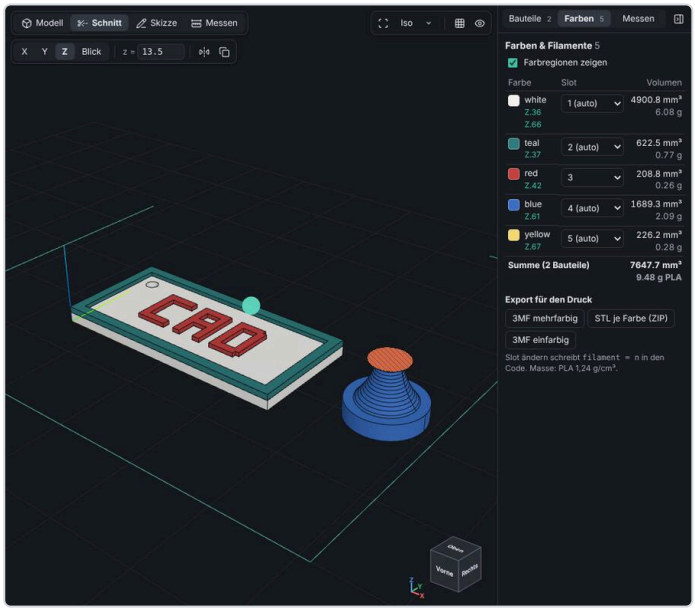
Before: model (mehrfarbig.scad), light



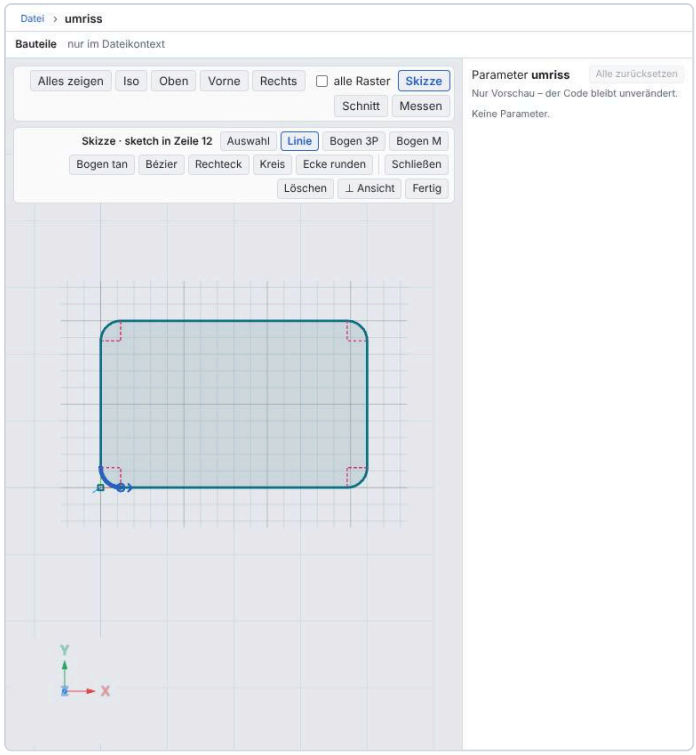
After: model (mehrfarbig.scad), light



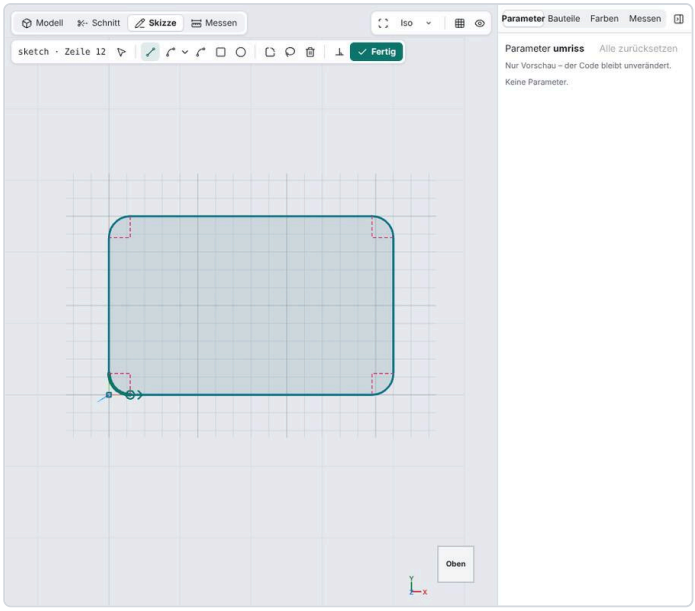
Before: section z, dark



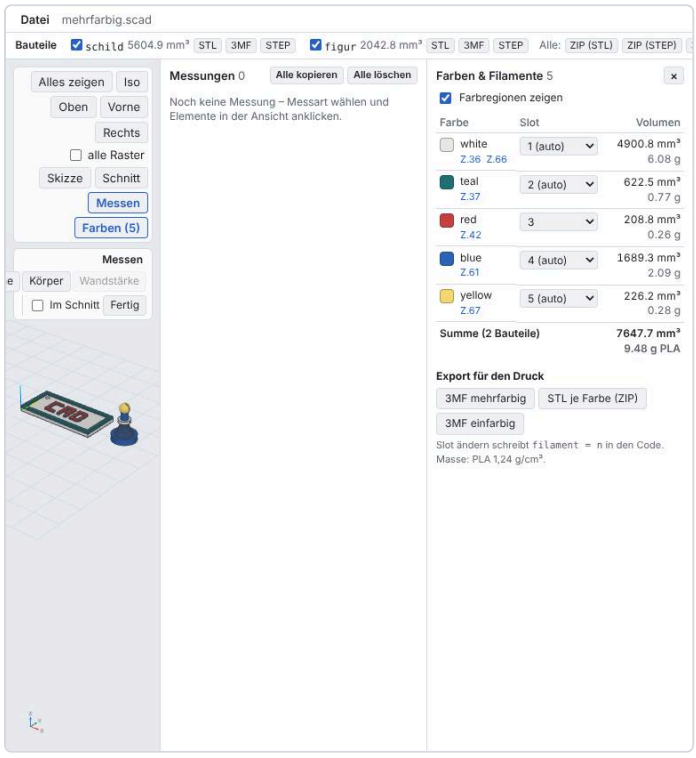
After: section z, dark



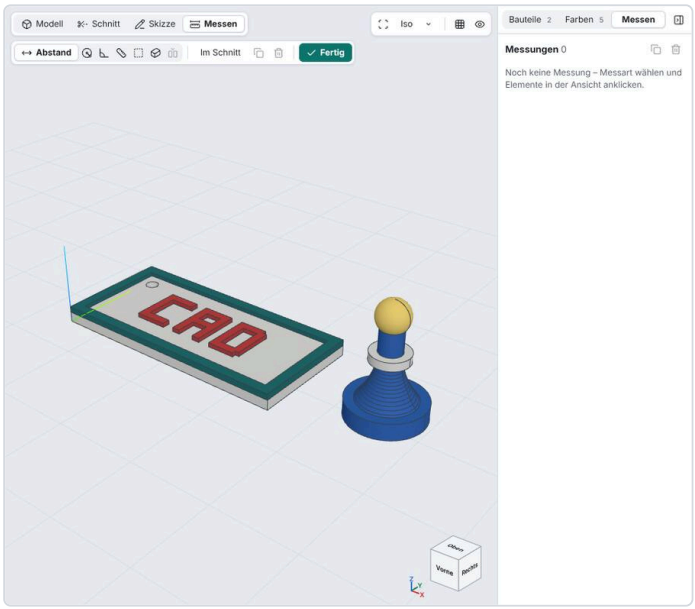
Before: sketch (skizze_deckel.scad, tool line), light



After: sketch (skizze_deckel.scad, tool line), light



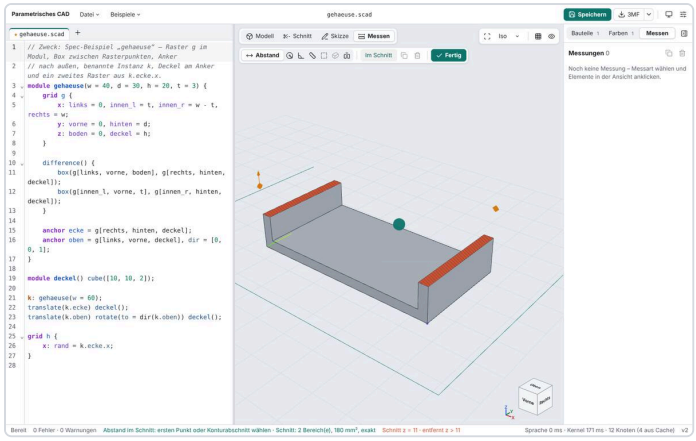
Before: measure, light



After: measure, light

U04 Mode

Model, Section, Sketch, Measure as a segmented control: always exactly one active. The mode's context bar sits at the top left below the toolbar. Measure "In section" turns the section on; after "Done" the view is back in Section mode. Arrow keys switch the mode, Esc exits Sketch and Measure.



measuring in the section, list in the "Measure" tab

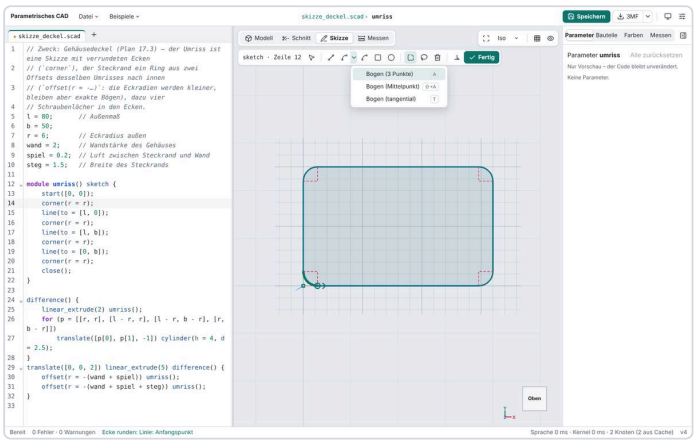
- exactly one aria-checked per state
- context bar below the toolbar, left-aligned
- arrow keys in the segmented control

U05 Sketch context bar

Icon groups Select | Line, Arc ▾, Bézier, Rectangle, Circle | Round corner, Close, Delete | ⊥ View and "Done" as primary button. Tooltips with keyboard shortcut; V, L, A, Shift+A, T, B, R, C, F pick the tools, the Arc menu the arc kind.



sketch context bar



arc kind

- v → select
- l → line
- a → arc_through
- Shift+A → arc_center
- t → arc_tangent
- b → bezier
- r → rect
- c → circle
- f → corner

U06 View cube

Clickable cube at the bottom right instead of the axis triad, rotates with the camera; face center = view onto the face, border strip = edge (diagonal), corners = corner. Next to it x/y/z.



after clicking "Top"



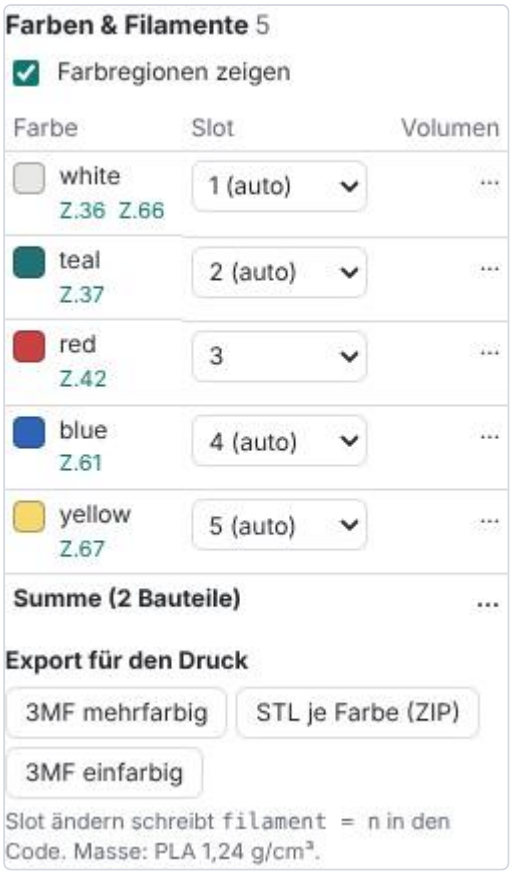
after clicking the edge Front/Right

- Top: view −z
- Front: view +y
- edge: view (−1, 1, 0)/√2

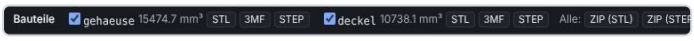
5. Side panel



Before: Colors & filaments, light



After: Colors & filaments, light



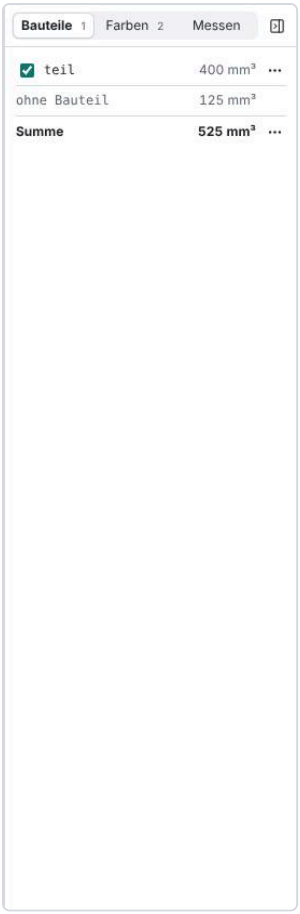
Before: parts (before a row above the view, after a tab), dark



After: parts (before a row above the view, after a tab), dark

U07 Side panel

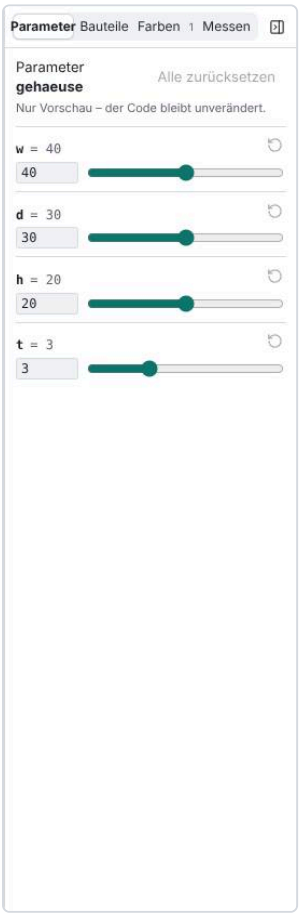
On the right a collapsible side panel with tabs Parts (visibility, volume, export in the ⋯ menu, "without part", total), Colors and Measure; in the module context additionally "Parameters" (the former parameter bar). Arrow keys switch the tabs. When collapsed the view gets the space, an icon button in the toolbar brings the panel back.



parts with "without part"



colors

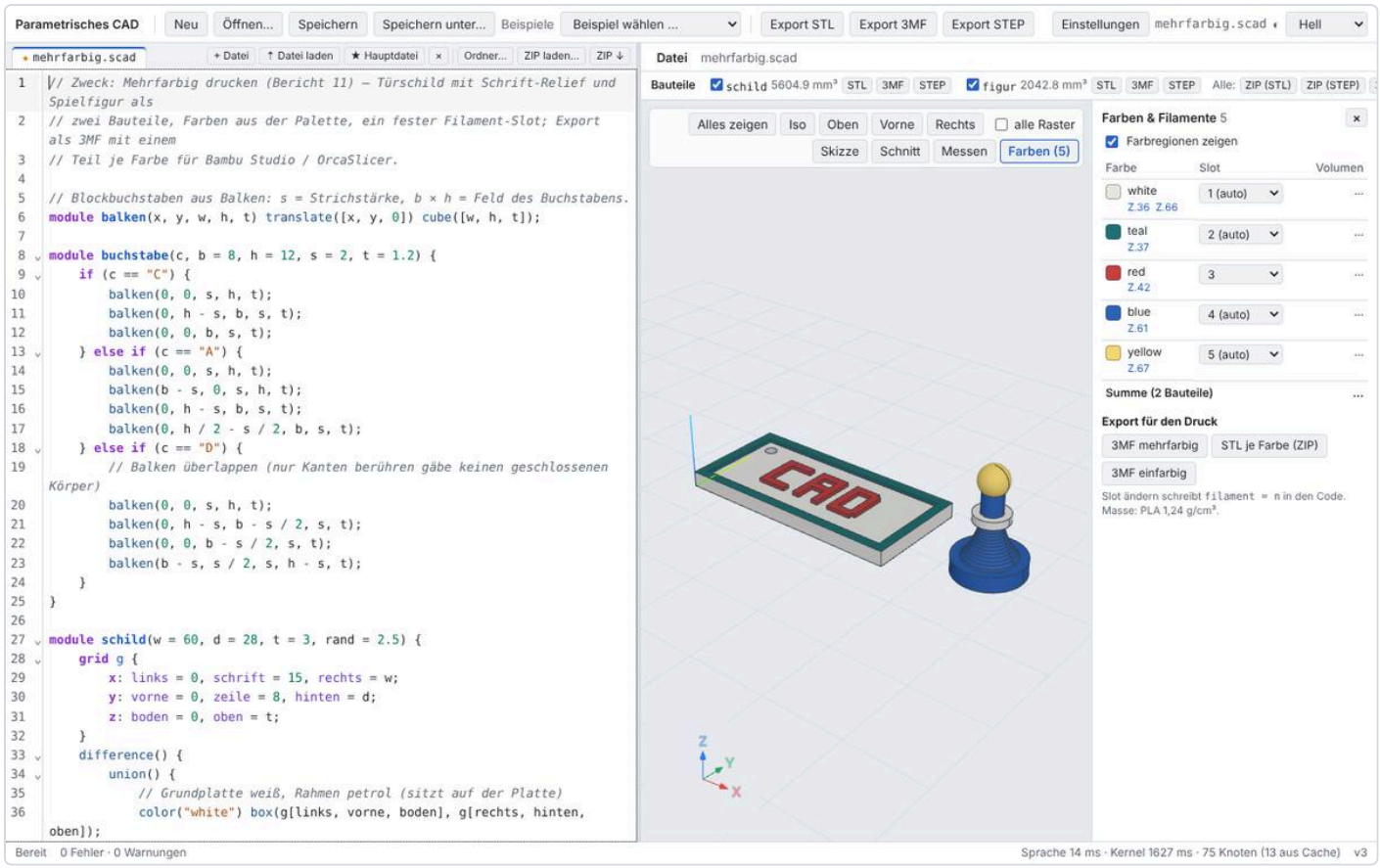


parameters in the module context

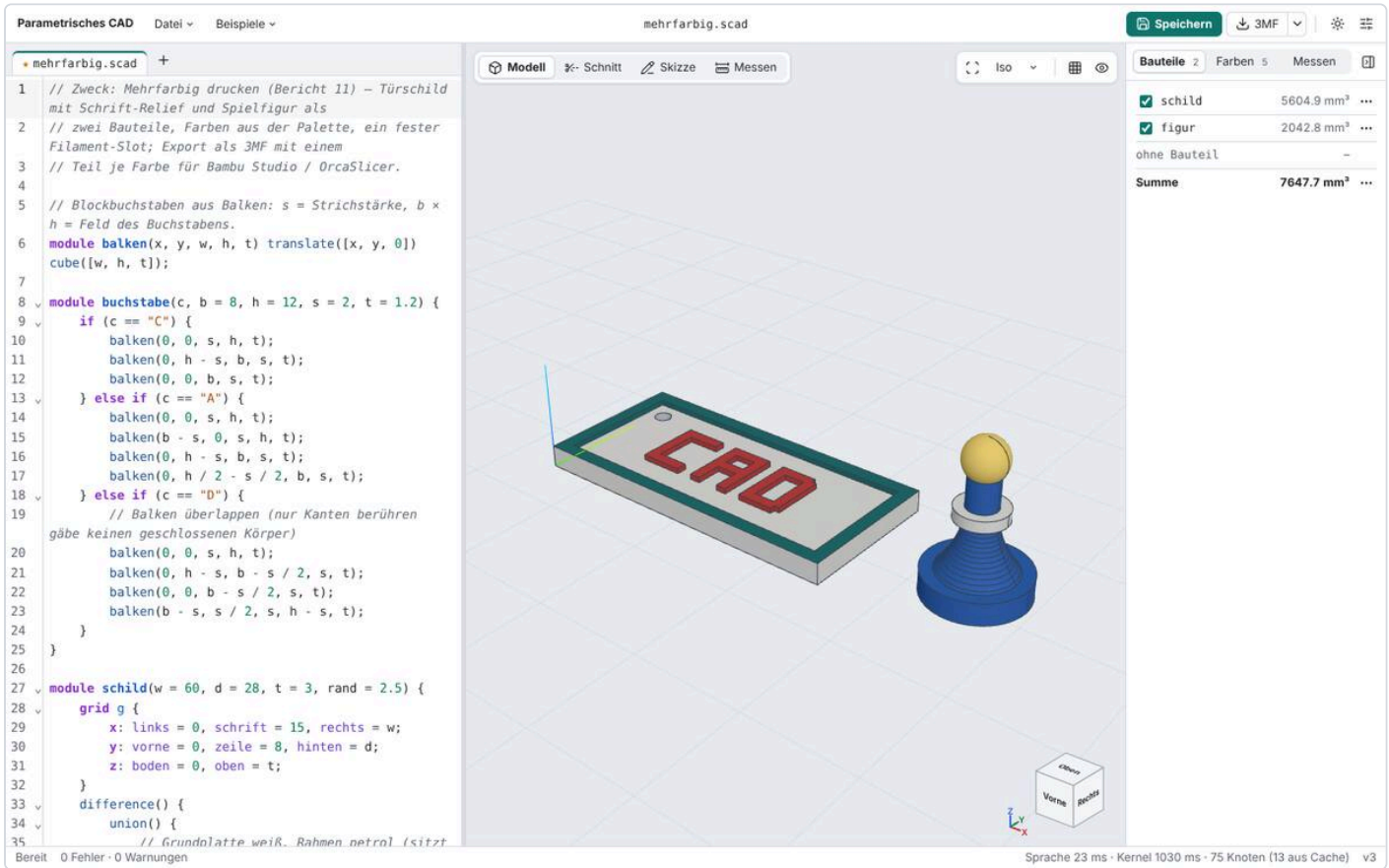
- teil 400 mm³, without part 125 mm³, total 525 mm³
- arrow keys switch tabs

- collapsed: view > 200 px wider

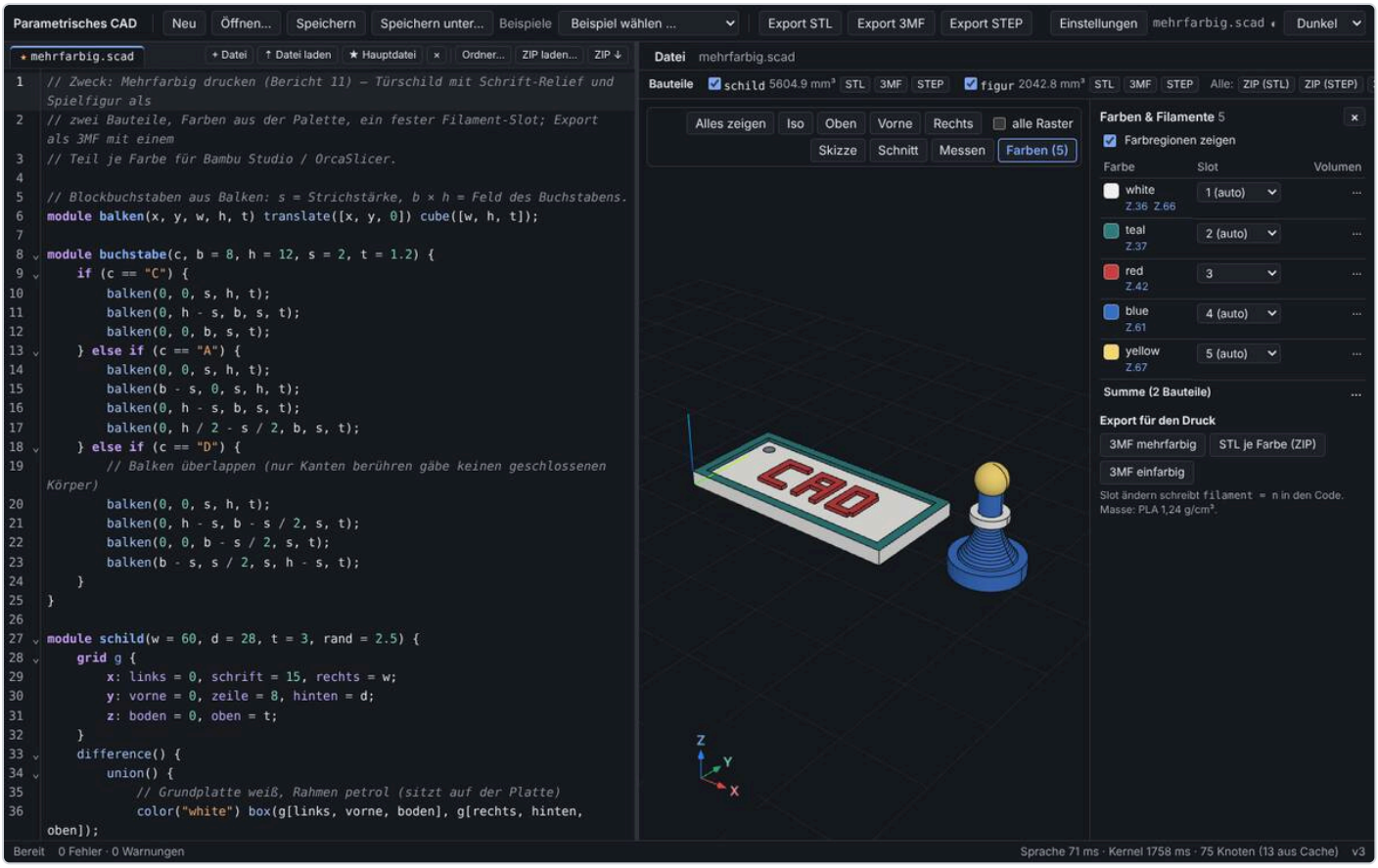
6. Light and dark



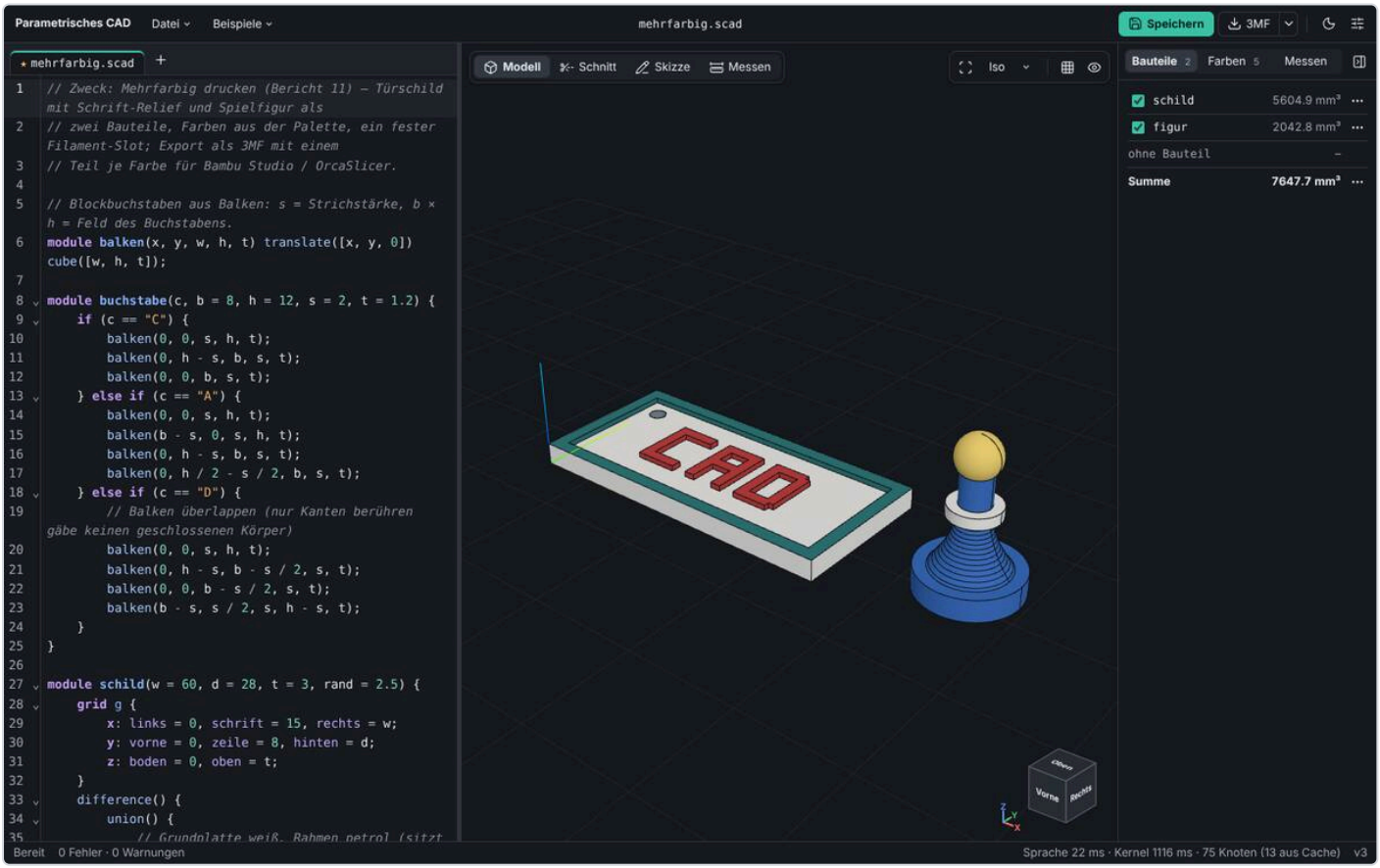
Before: full window 1440 px, light



After: full window 1440 px, light



Before: full window 1440 px, dark



After: full window 1440 px, dark

Accent and contrast (WCAG 2, `contrast.test.ts`)

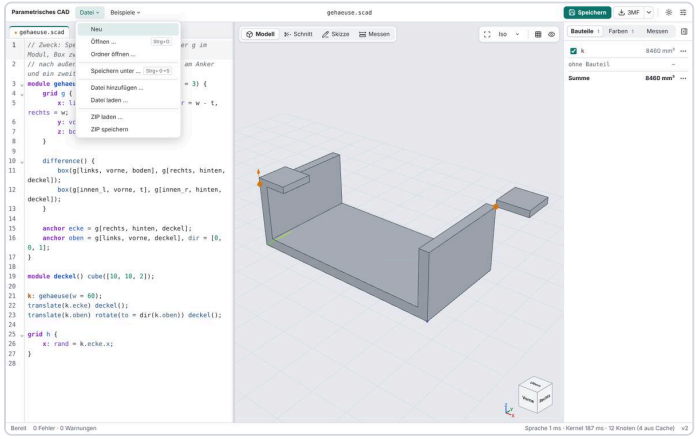
The token values are checked in `web/src/logic/contrast.test.ts` (text ≥ 4.5:1, controls ≥ 3:1); newly added are the pairs for buttons, segmented controls, menus, the focus ring and the view cube.

Token	light	dark
`--accent`	`#0f766e`	`#3cc4a8`
`--accent-hover`	`#115e59`	`#63d6bf`
`--accent-soft`	`rgba(15, 118, 110, 0.12)`	`rgba(60, 196, 168, 0.16)`
`--accent-text`	`#0b5e57`	`#5fd4bb`
`--on-accent`	`#ffffff`	`#04201b`
`--focus`	`#0f766e`	`#3cc4a8`
`--view-accent`	`#0f766e`	`#62dcc2`
`--view-flat`	`#0f766e`	`#3cc4a8`
`--hover-line`	`rgba(15, 118, 110, 0.08)`	`rgba(60, 196, 168, 0.08)`
`--viewhover-line`	`rgba(15, 118, 110, 0.2)`	`rgba(60, 196, 168, 0.2)`

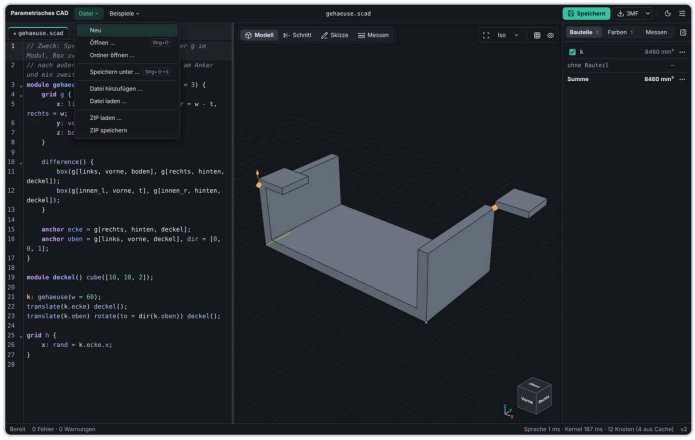
Pair	Use	Minimum	light	dark
`--accent/--panel`	accent as text (links, path) on panel	4.5	5.47	7.94
`--accent/--bg`	accent on background	4.5	5.11	8.48
`--on-accent/--accent`	text on primary button	4.5	5.47	7.86
`--on-accent/--accent-hover`	text on primary button (hover)	4.5	7.58	9.69
`--accent-text/--accent-soft`	active toggle, menu check mark	4.5	6.45	7.11
`--text/--accent-soft`	menu item with focus	4.5	13.36	10.55
`--text/--seg-on`	active segment (mode, tab)	4.5	15.80	9.41
`--text/--ctl-hover`	hover of quiet buttons	4.5	14.08	11.68
`--focus/--panel`	focus ring on panel	3.0	5.47	7.94
`--focus/--view-bg`	focus ring over the 3D view	3.0	4.54	8.11
`--view-accent/--view-bg`	view accent (hover, selection)	3.0	4.54	10.53
`--view-accent/--view-model`	accent on the model	2.0	3.32	2.11
`--text/--vc-face`	view cube label	4.5	14.58	11.07
`--axis-x/--view-bg`	x axis at the cube	3.0	4.02	6.37

U09 Themes

Accent teal (light #0f766e, dark #3cc4a8) for primary buttons, active toggles, menu highlight and focus ring; contrast measured on the rendered button (computed styles).



light, File menu open



dark, File menu open

- light save: 5.47
- light mode: 15.80

- light status: 5.80
- dark save: 7.86
- dark mode: 9.41
- dark status: 7.37

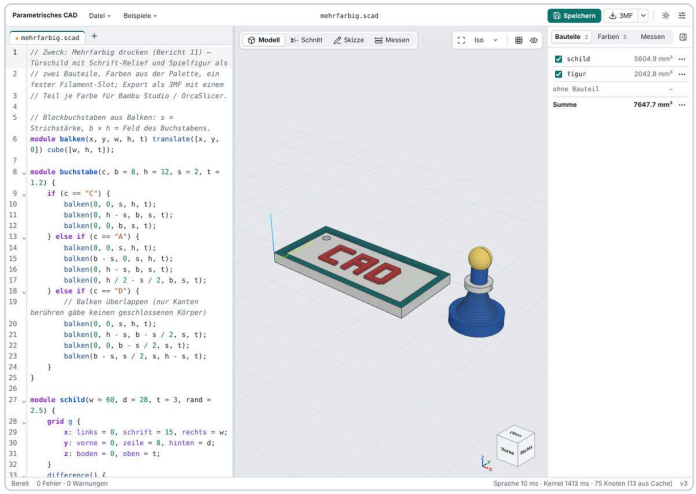
7. Widths 1280, 1440 and 1920 px

U01 measures every bar per width, theme and mode: `scrollWidth ≤ clientWidth` (nothing clipped or scrolled), no control extends beyond its bar or the window, every bar single-line, mode and camera bars do not overlap. For narrower views (small windows, dragged divider) the mode and measurement-kind segmented controls show icons only (container queries).

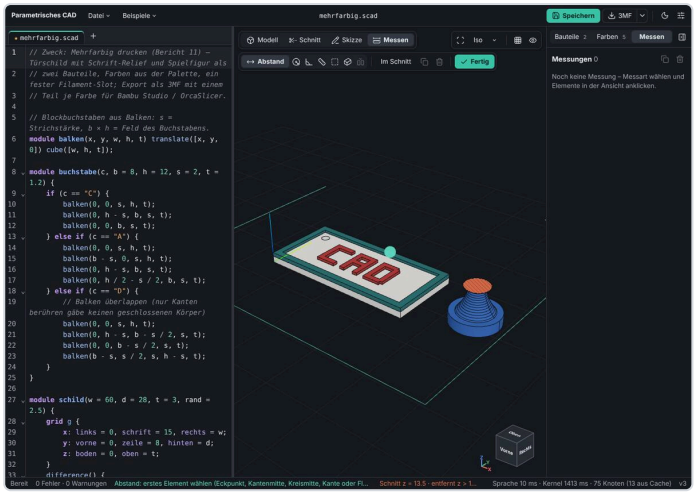
State (width, theme, mode)	max. scrollWidth − clientWidth	protruding elements	header bar px	gap mode–camera px	context bar px
1280 light Model	≤ 0	0	40	64	–
1280 light Section	≤ 0	0	40	64	316 × 34
1280 light Measure	≤ 0	0	40	64	481 × 34
1440 light Model	≤ 0	0	40	172	–
1440 light Section	≤ 0	0	40	171	316 × 34
1440 light Measure	≤ 0	0	40	171	481 × 34
1920 light Model	≤ 0	0	40	493	–
1920 light Section	≤ 0	0	40	493	315 × 34
1920 light Measure	≤ 0	0	40	493	831 × 34
1280 dark Model	≤ 0	0	40	64	–
1280 dark Section	≤ 0	0	40	64	316 × 34
1280 dark Measure	≤ 0	0	40	64	481 × 34
1440 dark Model	≤ 0	0	40	172	–
1440 dark Section	≤ 0	0	40	171	316 × 34
1440 dark Measure	≤ 0	0	40	171	481 × 34
1920 dark Model	≤ 0	0	40	493	–
1920 dark Section	≤ 0	0	40	493	315 × 34
1920 dark Measure	≤ 0	0	40	493	831 × 34
1280 light Sketch	≤ 0	0	40	65	552 × 34
1440 light Sketch	≤ 0	0	40	172	552 × 34
1920 light Sketch	≤ 0	0	40	493	552 × 34
1280 dark Sketch	≤ 0	0	40	65	552 × 34
1440 dark Sketch	≤ 0	0	40	172	552 × 34
1920 dark Sketch	≤ 0	0	40	493	552 × 34

U01 Widths and themes

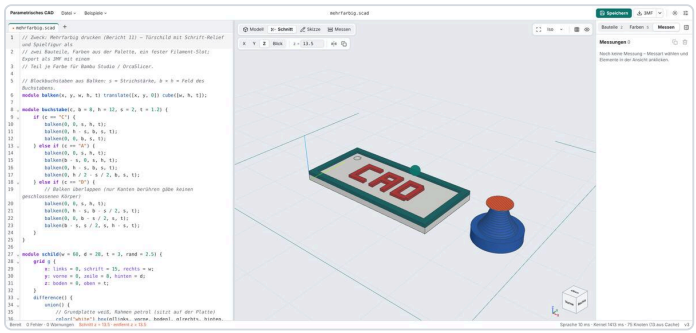
Per width (1280, 1440, 1920 px) and theme all bars in the modes Model, Section, Measure and Sketch: content not wider than the bar (scrollWidth ≤ clientWidth), no element protrudes, every bar single-line, mode and camera bars do not overlap, the window does not scroll horizontally.



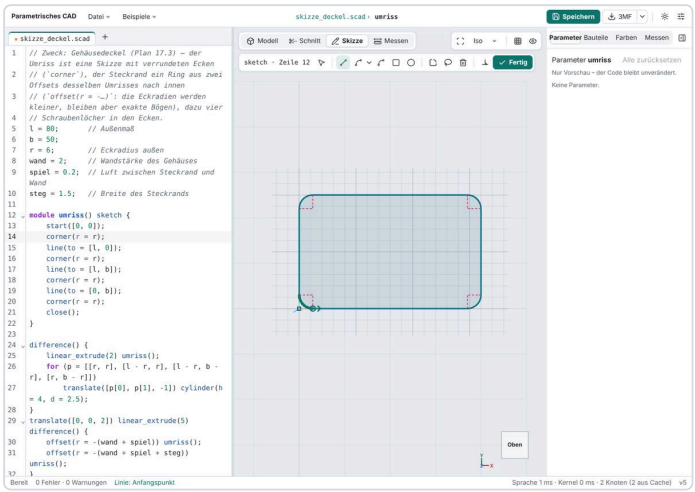
1280 px, light, Model



1280 px, dark, Measure



1920 px, light, Section

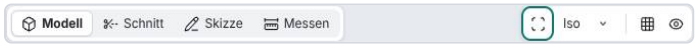


1280 px, light, Sketch

- 24 states checked, 0 findings

U08 Keyboard

Reachable with Tab: File, Examples, Save, Export (button and menu), Theme, Settings, tabs, "+", mode, Fit, View, grids, side panel tabs. Focus ring 2 px in the accent color.



focus ring in the toolbar

- focus order: File → Examples → Save → Export: 3MF multicolor → Export format → Color scheme → Settings → gehaeuse.scad → Add file → Source → Editor → Model → Fit → View → All grids → Display → Parts → Hide side panel …
- focus ring rgb(15, 118, 110)

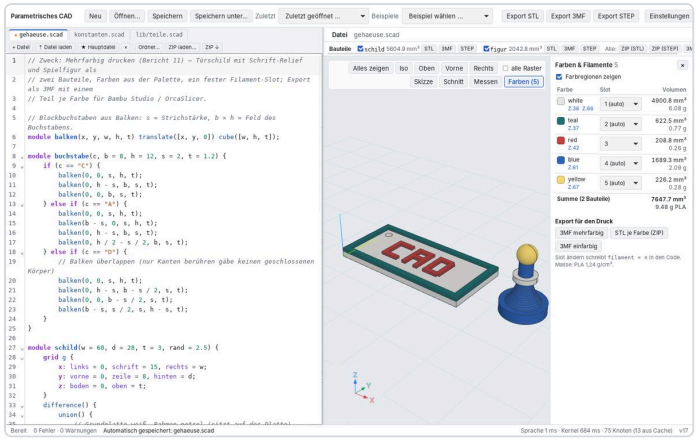
Keyboard shortcuts and keyboard

Area	Keys
File	Ctrl+N New (desktop app), Ctrl+O Open, Ctrl+S Save, Ctrl+⇧+S Save as, Ctrl+E export in the last chosen format
Menus	Enter/Space/↓ open, ↑↓ Home End move, → submenu, ← back, Esc closes (focus back to the button), Tab closes
Tabs	← → switch the file, Shift+F10 or context menu key: Set as main file, Close
Mode, segmented controls, tabs	← → (Home/End) select; exactly one value active
Sketch	V select, L line, A arc 3 points, ⇧A arc around center, T tangent, B Bézier, R rectangle, C circle, F round corner, Enter close, Del delete, N ⊥ view, Esc discard points / done

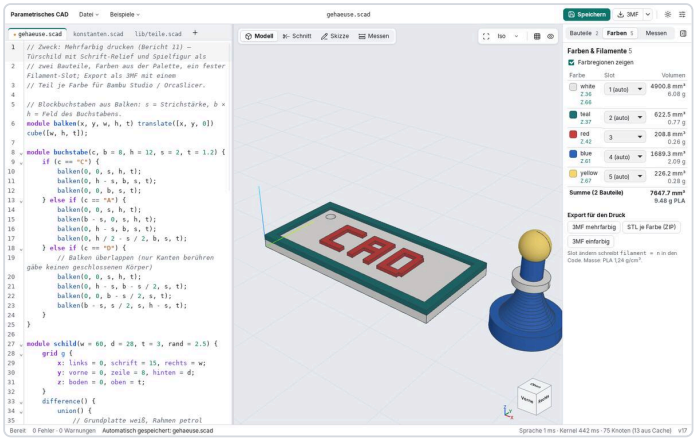
Measure	D distance, R radius, W angle, L length, F area, K body, T wall thickness, Del delete selected measurement, Esc discard selection / exit

8. Desktop app (Tauri)

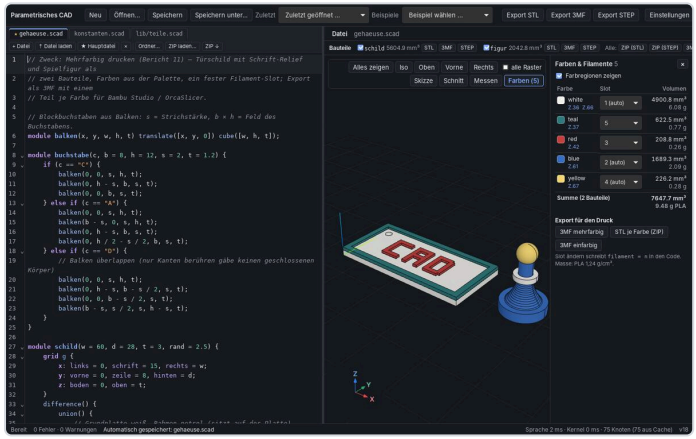
Same interface, embedded in cad (WebKitGTK under Xvfb, tauri-driver). Before: images of the old state from the same test sequence.



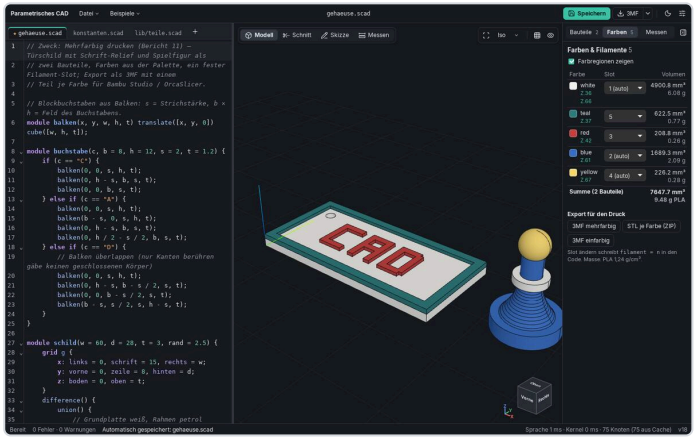
Before: T12 Colors & filaments (light)



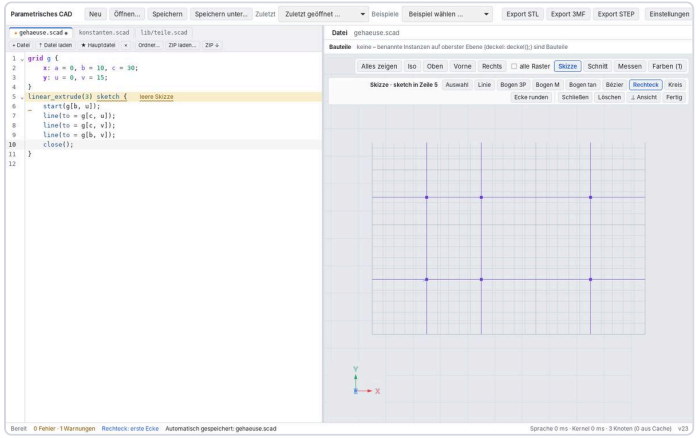
After: T12 Colors & filaments (light)



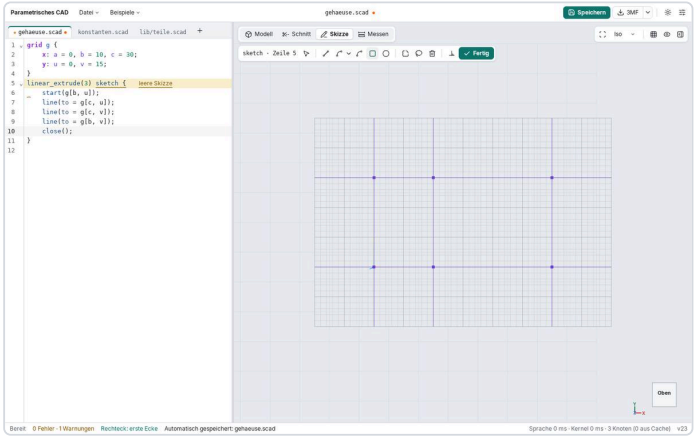
Before: T13 dark theme



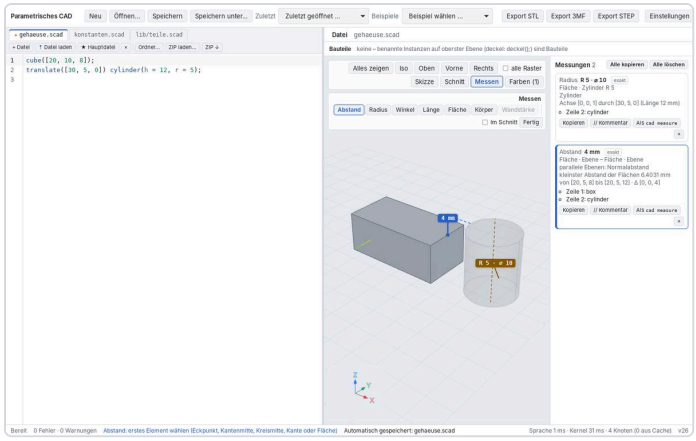
After: T13 dark theme



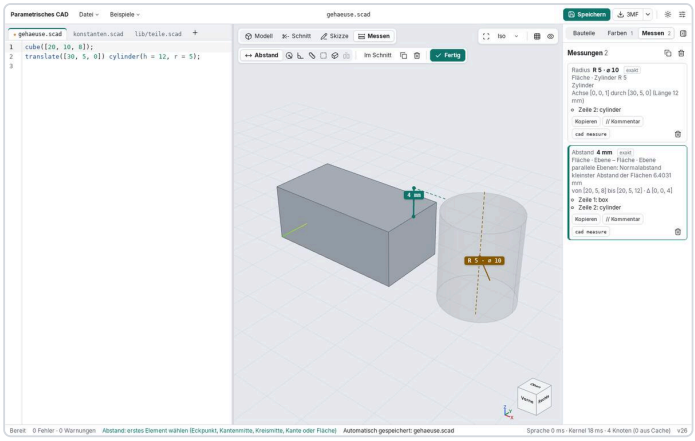
Before: T16 sketch, rectangle on grid points



After: T16 sketch, rectangle on grid points



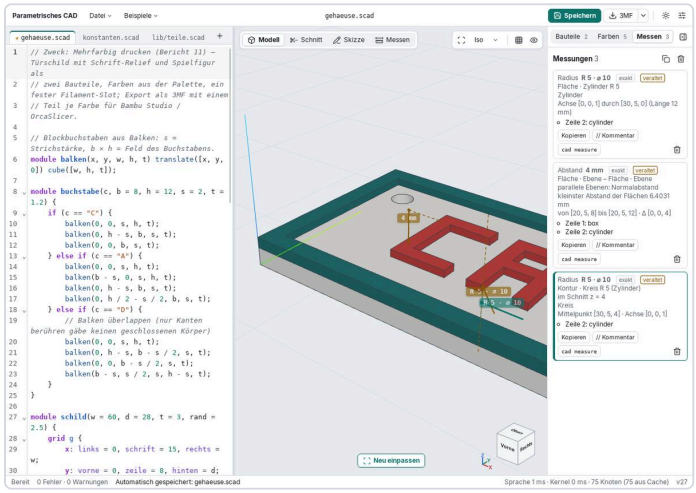
Before: T17 measure



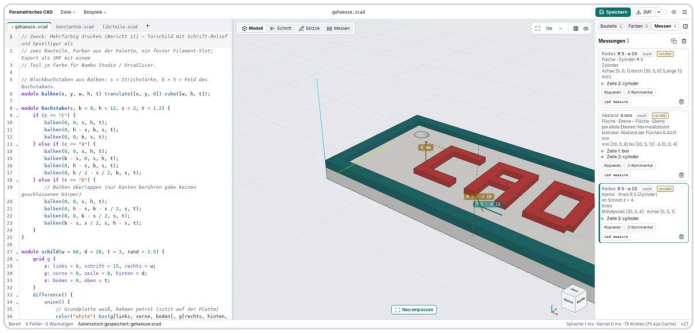
After: T17 measure

T18 UI rework in Tauri: bars without wrapping at 1280/1440/1920 (light/dark), menus, view cube, side panel

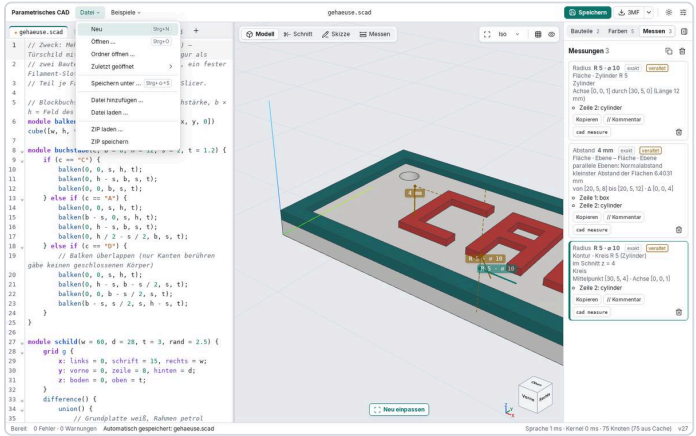
- menu closes on the second click
- export formats: 3MF multicolor, 3MF single color, STL, STL per color (ZIP), STEP, Per part
- menu closes on the second click
- export formats: 3MF multicolor, 3MF single color, STL, STL per color (ZIP), STEP, Per part
- no clipped or wrapped bars (12 states, window widths 1280/1440/1920 px)
- view cube "Top": view direction 0.000, 0.000, -1.000



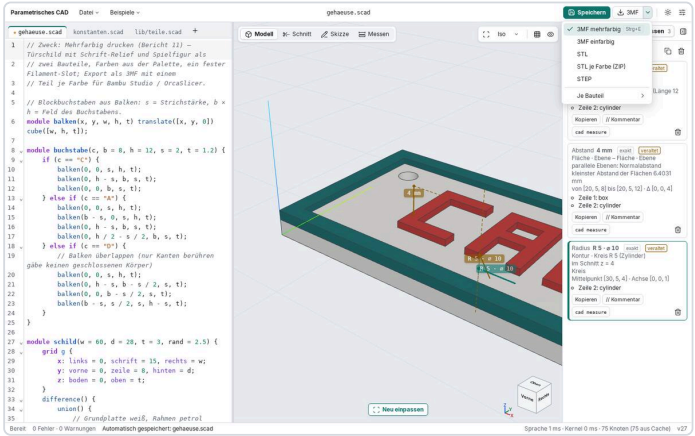
1280 px, light



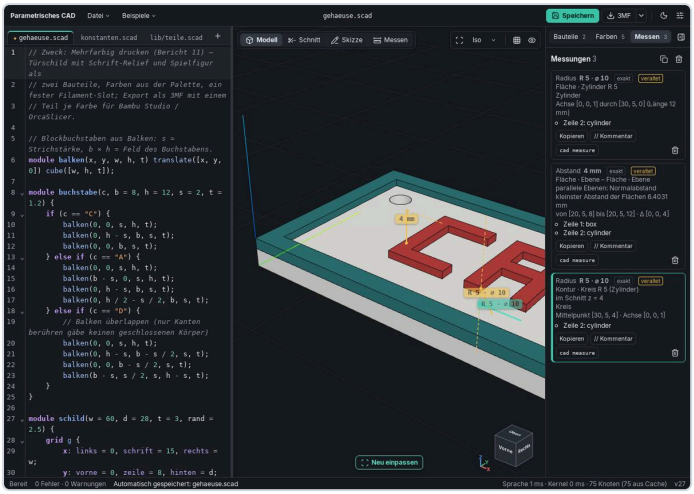
1920 px, light



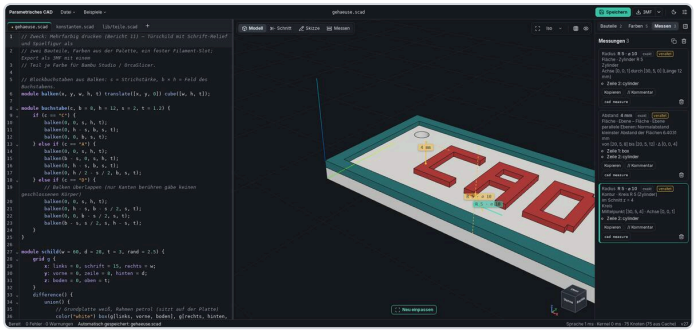
File menu, light



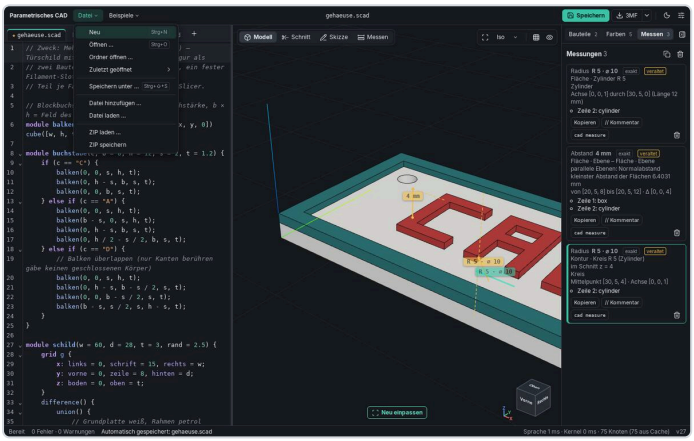
Export menu, light



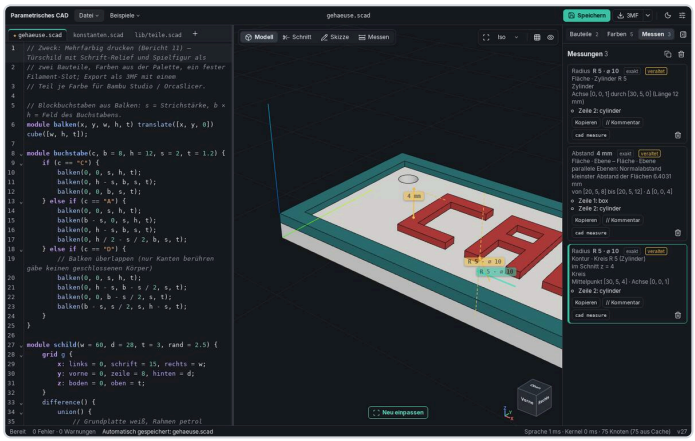
1280 px, dark



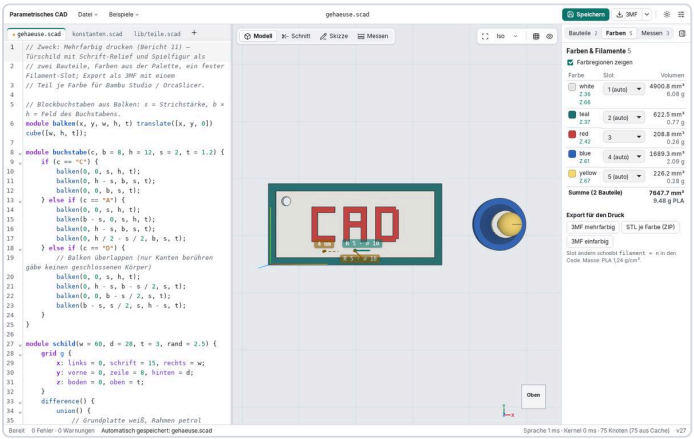
1920 px, dark



File menu, dark



Export menu, dark



view cube "Top", Colors tab

All Tauri scenarios

No.	Scenario	Result	Duration
T01	App starts with a file, scene appears (native kernel)	passed	3.3 s
T02	External change without unsaved changes: silently reloaded, undo as a separate step	passed	0.9 s
T03	Git markers in the gutter (new, changed, deleted)	passed	1.1 s
T04	External change with unsaved changes: notice bar, compare, keep mine, save without loop	passed	2.7 s
T05	Deleted and renamed file	passed	1.3 s
T06	Agent flow: change file via shell, `cad check`, UI follows (image sequence)	passed	1.3 s
T07	`cad ui status` / `show` / `screenshot` against the running app	passed	1.2 s

T08	Section view in Tauri (cap, contour, dragging, CLI call)	passed	5.6 s
T09	Watchdog in Tauri: hanging kernel → engine process replaced, node blocked	passed	7.5 s
T10	Folder as project, autosave	passed	3.5 s
T11	Keystroke → scene natively (measurement)	passed	12.2 s
T12	Colors & filaments in Tauri: list, change slot, 3MF multicolor (native kernel)	passed	4.5 s
T13	Dark theme in Tauri: contrast (computed styles), view, editor	passed	0.7 s
T15	Report 12 in Tauri: color regions per instance, filament() table, object "without part", 3MF without color group	passed	1.0 s
T16	Sketch editor in Tauri: drawing mode, rectangle on grid points, snap indicator, undo (TauriTransport, native kernel)	passed	1.9 s
T17	Measuring in Tauri: radius, distance of parallel faces, radius in the section (TauriTransport, native kernel)	passed	2.4 s
T18	UI rework in Tauri: bars without wrapping at 1280/1440/1920 (light/dark), menus, view cube, side panel	passed	12.0 s
T14	Theme "System" in Tauri follows the GTK color scheme (prefers-color-scheme)	passed	4.8 s

Before run (old state): 17 / 17 passed.

9. Arch package: app ID under Wayland

Under Wayland the taskbar maps a window to the desktop file of the same name via its `app_id` (KDE, GNOME). Tauri does not set the app ID itself (`enableGTKAppId` is off because a registered GApplication would hand a second instance over to the first); GTK therefore took the program name `cad`, and the desktop file was named `controller-cad.desktop`. Now:

- `crates/cad-app` sets the GLib program name to the `identifier` from `tauri.conf.json` (`dev.controller-cad.desktop`) before GTK starts; GTK uses it as Wayland `app_id` and as X11 `WM_CLASS`. Configuration and WebView data stay under the same identifier (no migration).
- The desktop file is named after the app ID: `packaging/linux/dev.controller-cad.desktop.desktop`, `StartupWMClass=dev.controller-cad.desktop`; `make install` derives the name from `tauri.conf.json`, `make uninstall` also removes the old name; `packaging/arch/smoke-test.sh` checks the file and `WM_CLASS`.

Before (the old state copied as `cad-old`: the program name is the file name of the binary):

```
Fenster (xid, Titel, WM_CLASS res_name, res_class) unter Xvfb:
  0x400001  'cad-old'                                'cad-old'  'Cad-old'
  0x400003  ''                                       'cad-old'  'Cad-old'
Desktop-Datei packaging/linux/dev.controller-cad.desktop.desktop: vorhanden
WM_CLASS = App-ID = Name der Desktop-Datei: NEIN
```

After (`tools/report/report20-wmclass.py`, libX11 via ctypes under Xvfb):

```
Fenster (xid, Titel, WM_CLASS res_name, res_class) unter Xvfb:
  0x200001  'dev.controller-cad.desktop'             'dev.controller-cad.desktop'  'Dev.controller-cad.desktop'
  0x200003  ''                                       'dev.controller-cad.desktop'  'Dev.controller-cad.desktop'
Desktop-Datei packaging/linux/dev.controller-cad.desktop.desktop: vorhanden
WM_CLASS = App-ID = Name der Desktop-Datei: ja
```

10. Test runs

Run	Result
Playwright, full suite (15 files)	140 passed, 0 failed, 0 skipped
Tauri E2E (tauri-driver, WebKitWebDriver, Xvfb)	18 / 18
Vitest	238 passed
tsc --noEmit	no errors
Rust cad-app (program name): cargo test / clippy -D warnings / fmt --check	1 passed, 0 failed · clippy ok · fmt ok

The Playwright tests find controls by role and name (`getByRole("radio", { name: "Schnitt" })`, menu items via `menuitem`/`menuitemradio`, helpers in `web/e2e/helpers.ts`); test IDs remain only on displays (status line, labels in the view, list rows). The Tauri scenarios select via `[role=…][aria-label=…]`.

11. Deviations from the concept and limitations

- **Renaming** a tab does not exist (did not exist before); the context menu has "Set as main file" and "Close".
- **Parameter bar**: the concept does not mention it; it is now the "Parameters" tab of the side panel, which appears and is selected in the module context (before a separate column that narrowed the view further). On leaving the module the panel returns to the previous tab.
- **Path bar**: the path into the module context sits centered in the header bar next to the file name (click jumps back), instead of in the removed row "File …".
- **X-ray/ghost** in the view area: the UI had no such toggle (only the CLI `cad render --xray`); the other display options are in the "Display" menu (eye icon): All grids, Show color regions (also in the Colors tab), Side panel.
- **Measure** additionally has "Done" (like Sketch); copy/delete act on the selected measurement, otherwise the last one. "In section" turns the section on; after "Done" the view is in Section mode (the section is view state, Measure and Sketch leave it in place).
- **Default split** editor : view 33 : 67 (before 46 : 54; in the concept draft 30 %), so that toolbar and context bars stay single-line at 1280 px with the side panel open and the mode appears with text; the divider can still be dragged, narrow views show mode and measurement kind as icons.
- **Ctrl+N** is intercepted only by the desktop app (Chromium reserves the key for a new window); in the browser no shortcut is shown on the menu item.
- **View cube**: face centers are buttons with names ("View Top"), edges and corners mouse only; via keyboard there is the "View ▾" menu. The cube is an orthographic SVG projection (no perspective), a click jumps without animation.
- The colors of the model palette, the git markers and syntax highlighting are unchanged; `--view-profile` (profiles, petrol) is close in color to the new accent.
- Wayland itself cannot be checked in the container; what is checked is the program name (X11 `WM_CLASS` under Xvfb), which GTK also uses as `app_id`.